

IMPACT AND BENEFITS



Impact :

1. Better Future Security
2. Early Attack Detection
3. More Reliable Verification
4. Easy to Understand
5. Useful for Future Quantum Networks
6. No AI/ML Dependency

Benefits :

1. Better security against quantum-era attacks.
2. Detects forgery and impersonation.
3. Detects replay attacks.
4. Reduces false alarms by considering normal quantum randomness.
5. Faster verification through early decision-making.
6. Explains why a signature is suspicious, rather than only saying "invalid."

```
quantum-qds-prototype
├── .gitignore
├── README.md
├── run_backend.sh
├── run_frontend.sh
├── backend
│   ├── requirements.txt
│   ├── api
│   │   ├── __init__.py
│   │   └── main.py
│   ├── detection
│   │   ├── init.py
│   │   ├── attacks.py
│   │   ├── engine.py
│   │   ├── models.py
│   │   └── registry.py
│   └── quantum
│       ├── init.py
│       ├── constants.py
│       ├── disturbance.py
│       ├── fidelity.py
│       ├── geometry.py
│       ├── measurement.py
│       ├── pauli_measurement.py
│       ├── shot_experiment.py
│       ├── state.py
│       ├── state_registry.py
│       ├── statistical_detector.py
│       ├── teleportation.py
│       └── test_state.py
├── tests
│   ├── init.py
│   ├── test_api_logic.py
│   └── test_quantum.py
└── frontend
    ├── index.html
    ├── package.json
    ├── vite.config.js
    └── src
        ├── main.jsx
        └── styles.css
```